DP Biology – A2.2 (HL extension)

Cell origins and specialization

- What do all cells share, and how do they differ?
- How do we use microscopes to study cell structure?

HL statements A2.2.12–14. Textbook pp. 82–87.

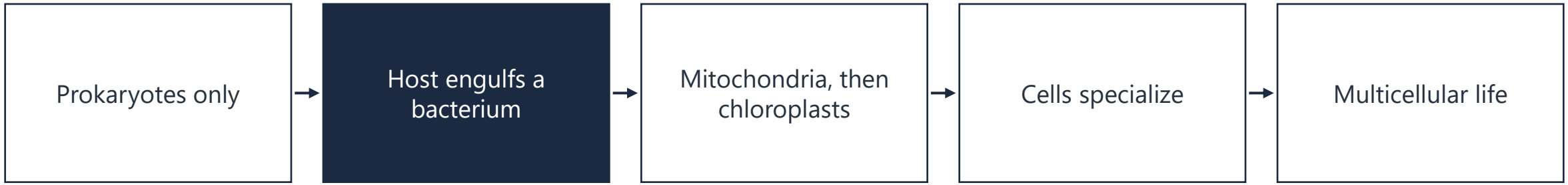


Image: Paramecium bursaria with green algae living inside it. Picturepest, CC BY 2.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	A2.2.12	Explain the endosymbiotic origin of mitochondria and chloroplasts, and evaluate the evidence for it
2	A2.2.13–14	Explain differentiation as a result of gene expression, and outline the evolution of multicellularity

The big picture: partnerships and teamwork



Eukaryotic cells arose from **partnerships** of cells. Later, cells in one body **specialized** and multicellular life evolved several times.

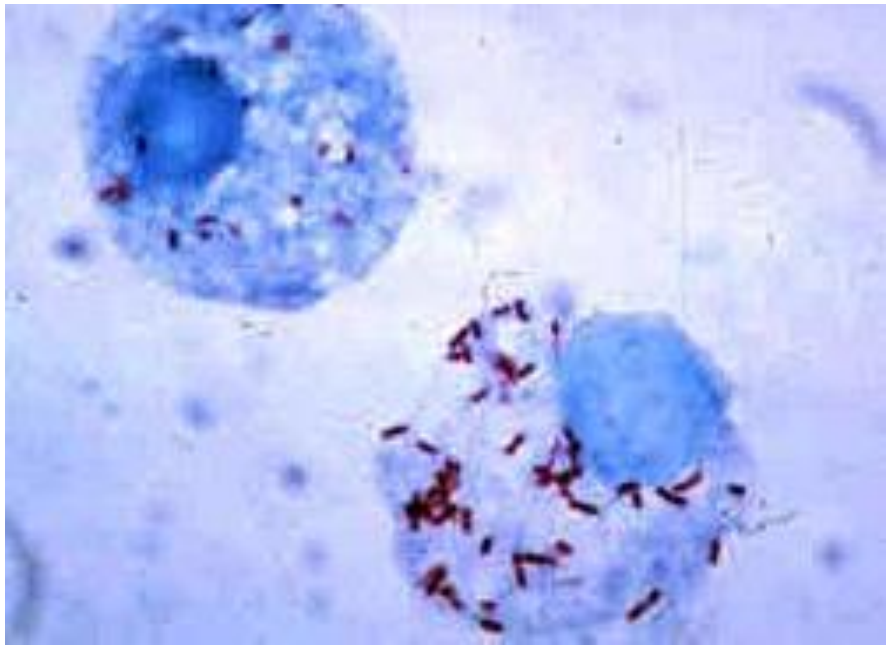
Big question

Why do your mitochondria have their own DNA?

Where did eukaryotic cells come from?

A2.2.12

Evidence	Approximate age
Oldest fossil prokaryotes	About 3500 million years
Oldest fossil eukaryotes	About 1600–1800 million years
Oldest multicellular eukaryotes	About 1000–1600 million years



Rickettsia (red rods) inside host cells. CDC, public domain.

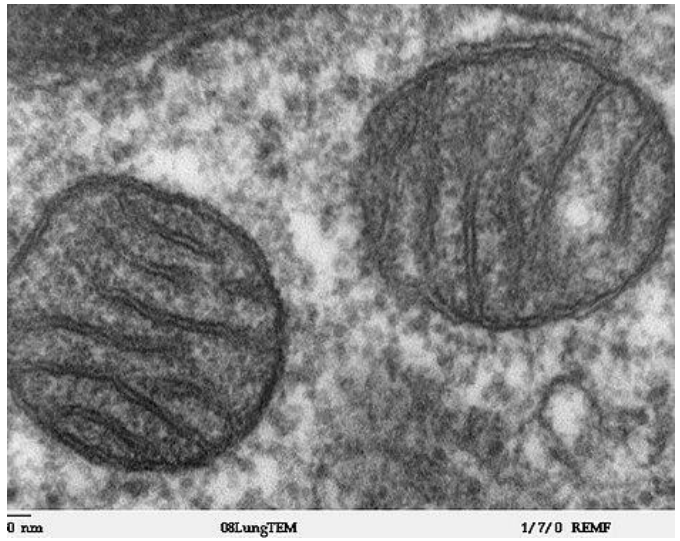
Key idea

Mitochondrial DNA is closest to alphaproteobacteria, the group that includes Rickettsia.

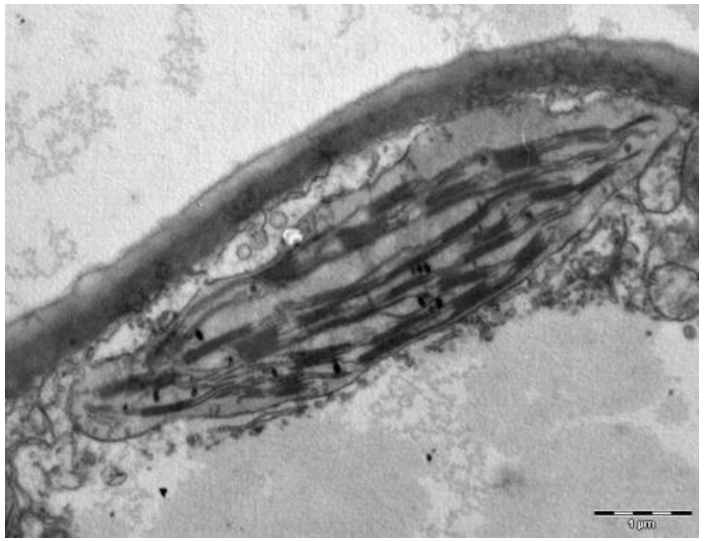
Endosymbiosis step by step

A2.2.12

Host engulfs an aerobic bacterium → Not digested: it supplies ATP → Both benefit (mutualism) → It becomes a mitochondrion → Later a cyanobacterium becomes a chloroplast



Mitochondria (TEM). Louisa Howard, public domain.



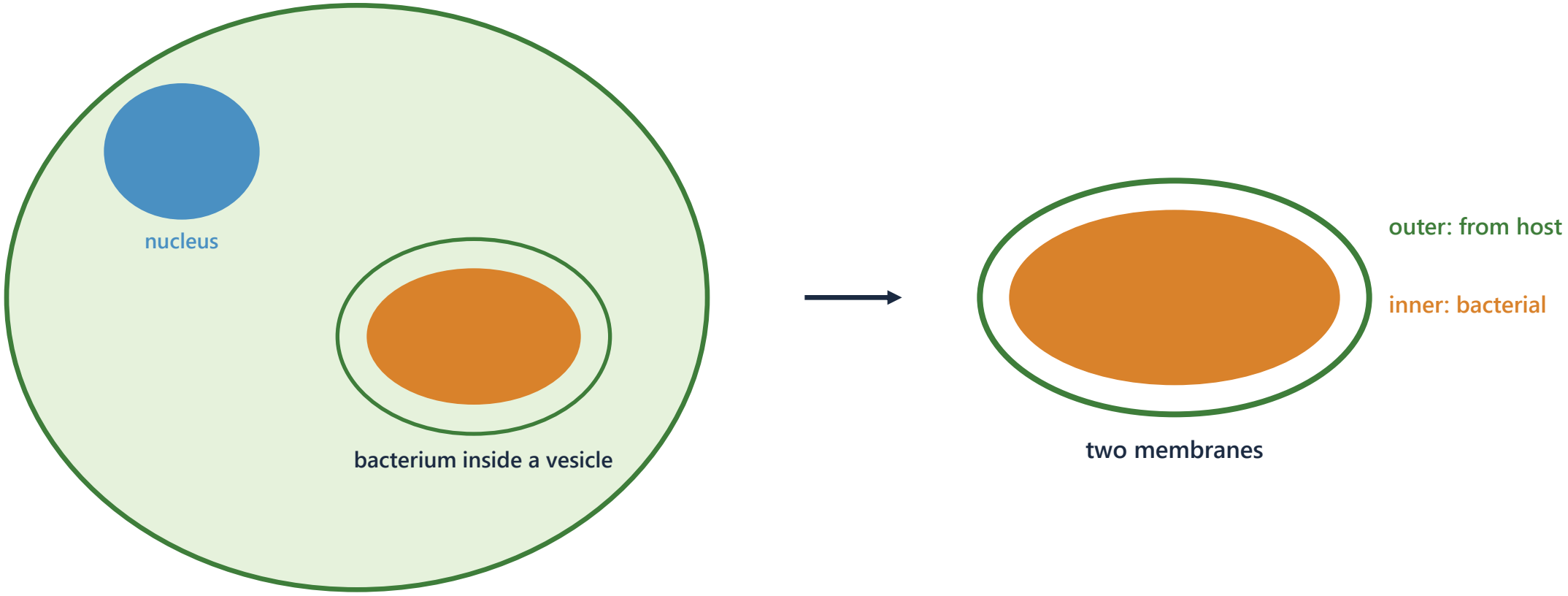
Chloroplast (TEM). and3k and caper437, CC BY-SA 3.0.

Common mistake

Mitochondria came first, so all eukaryotes have (or had) them; only some lineages later gained chloroplasts.

How the double membrane formed

A2.2.12



Prediction

The inner membrane should be bacteria-like: it is (cardiolipin, bacterial-type transport proteins).

Evidence for endosymbiosis

A2.2.12

Observation	Why it supports the theory
70S ribosomes	As in bacteria; eukaryotic cytoplasm has 80S
Naked circular DNA	Like a bacterial chromosome
Binary fission	Form only from existing organelles
Double membrane	Outer one from the host's vesicle
Bacterial size	About 1–10 μm
DNA sequences	Match alphaproteobacteria and cyanobacteria

Making predictions from the theory

A2.2.12



Paramecium bursaria. Picturepest, CC BY 2.0.

1. **Predict** whether antibiotics that block bacterial ribosomes affect mitochondria.
2. **Explain** why human mitochondrial DNA has only 37 genes.
3. **Discuss** how Paramecium bursaria, with algae inside, supports the theory.

Nature of science

A strong theory makes testable predictions; DNA sequencing later confirmed this one.

Working with data: comparing ribosomal RNA

A2.2.12

Similarity of rRNA sequences (%). Invented values; pattern matches real comparisons.

Compared with	Mitochondrial	Chloroplast
Alphaproteobacteria	78	52
Cyanobacteria	54	81
Plant nuclear rRNA	40	42

1. Identify each organelle's closest relative.
2. Explain how this supports endosymbiosis.
3. Why is the match not 100%?

Endosymbiosis happening now

A2.2.12

Paulinella

An amoeba holding a photosynthetic bacterium taken in about 100 million years ago. It has lost genes and cannot live alone.

Corals

Corals keep photosynthetic algae in their cells: sugars for shelter. Stress makes them expel the algae (bleaching).

Discuss

What does a partner lose, and gain, as it becomes an organelle?

Practice: endosymbiosis

A2.2.12

Outline

The steps that produced mitochondria.

Evaluate

Four pieces of evidence.

Explain

Why all eukaryotes have mitochondria but not all have chloroplasts.

Use

endocytosis, mutualism, 70S, binary fission

Check-in: class 1

A2.2.12

1. Outline how mitochondria arose.
2. Explain why mitochondria have two membranes.
3. Give four pieces of evidence for endosymbiosis.
4. State the closest living relatives of mitochondria and chloroplasts.
5. Explain why mitochondrial DNA is small.

How to use this

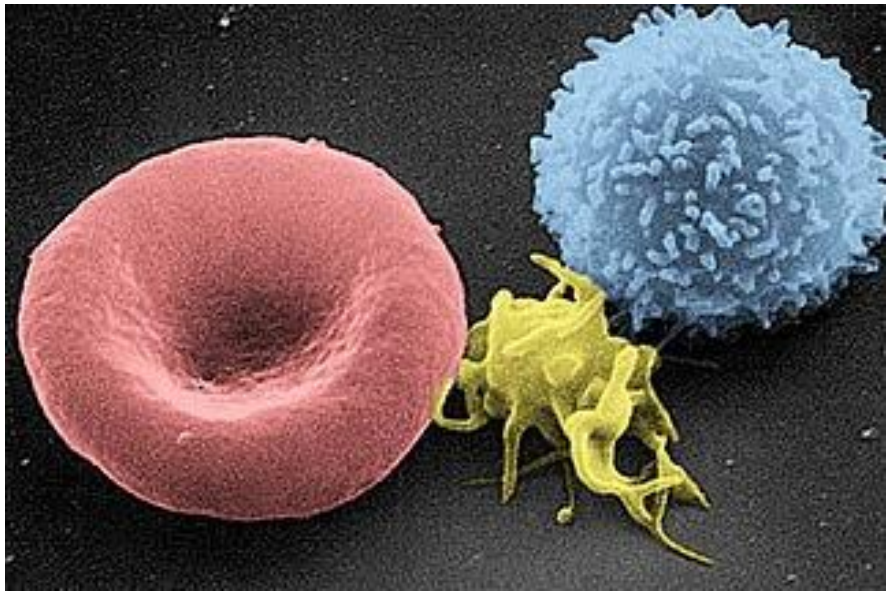
Answer without your notes, then check against slides 3–11.

Differentiation: same genes, different expression

A2.2.13

Every cell has the same **genome**, from one zygote by mitosis.

In **differentiation**, genes are switched on or off by signals and position, so each cell type has a different **proteome**.



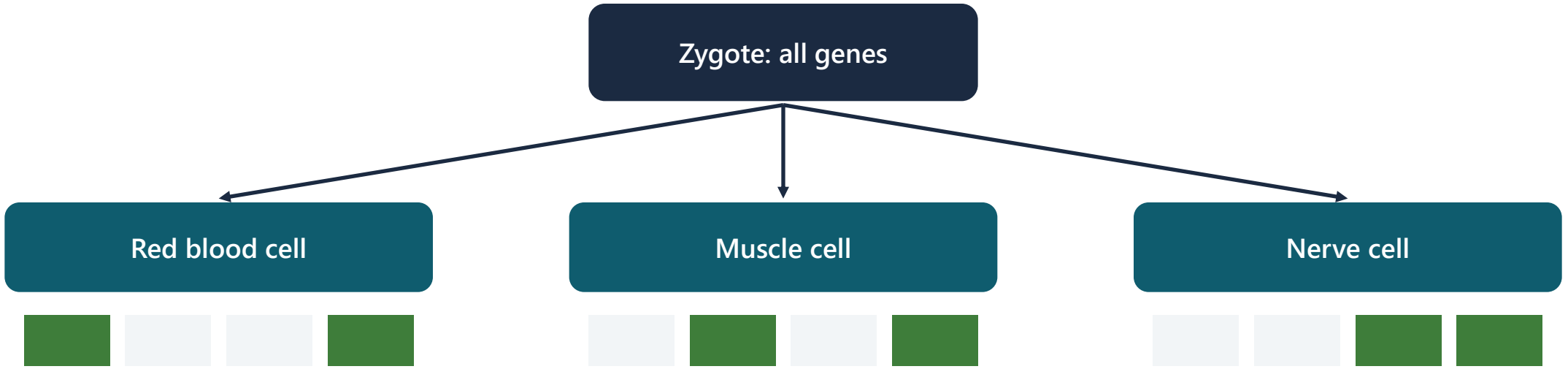
Blood cells (SEM). NCI, public domain.

Example

Developing red cells express haemoglobin genes; muscle cells express actin and myosin.

One genome, many cell types

A2.2.13



gene boxes: haemoglobin, myosin, ion channels, respiration; green = switched on

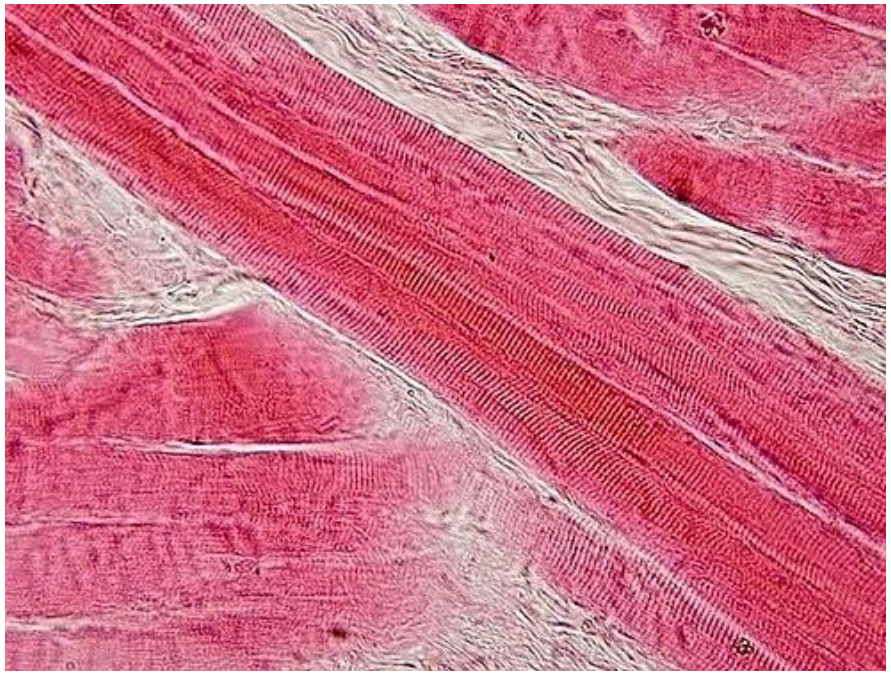
Key idea

Cells keep every gene but express only some. Housekeeping genes (respiration) are on in all cells.

From cells to tissues and organs

A2.2.13

Level	Example
Specialized cell	Muscle fibre
Tissue	Skeletal muscle
Organ	Small intestine
Organ system	Digestive system



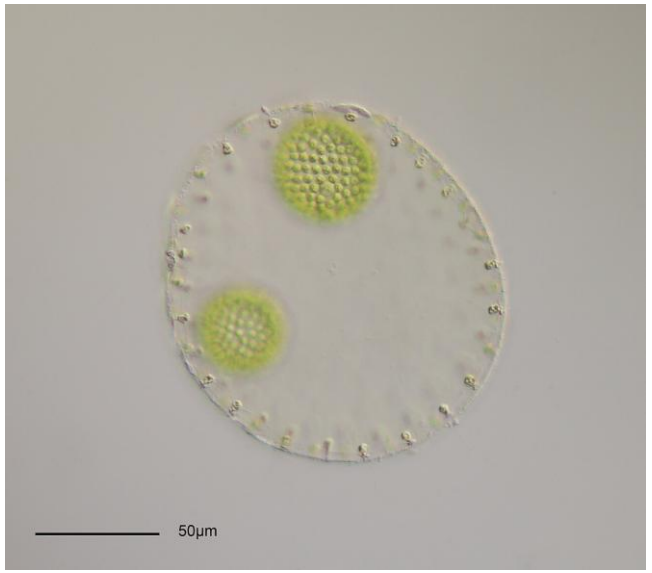
Skeletal muscle (light micrograph). Wikimedia Commons, CC BY-SA 3.0.

Cost

A nerve cell relies on blood and the heart for oxygen.

Multicellularity evolved many times

A2.2.14



Volvox colonies with daughter colonies inside.
Picturepest, CC BY 2.0.

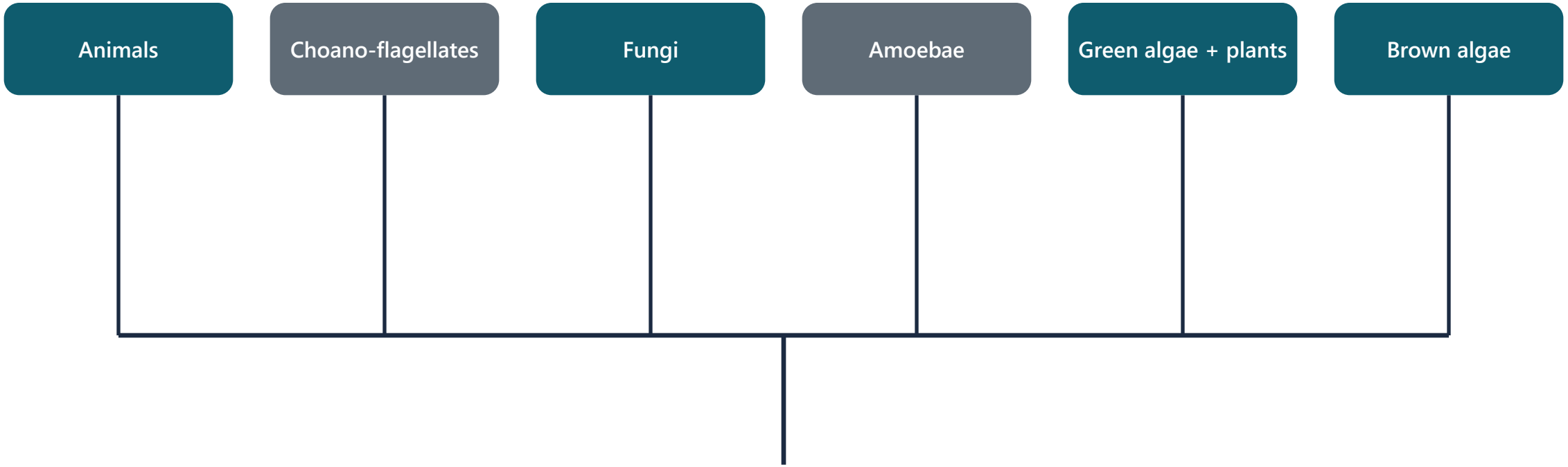
Animals, plants, many fungi and many algae are multicellular, but these groups are far apart on the tree of life: multicellularity arose **independently** many times.

Volvox: a ball of flagellated cells with a few reproductive cells.

Advantages	Requirements
Larger size; new niches	Cells stick together
Specialized cells work efficiently	Cells signal and differentiate

Many origins on the tree of life

A2.2.14



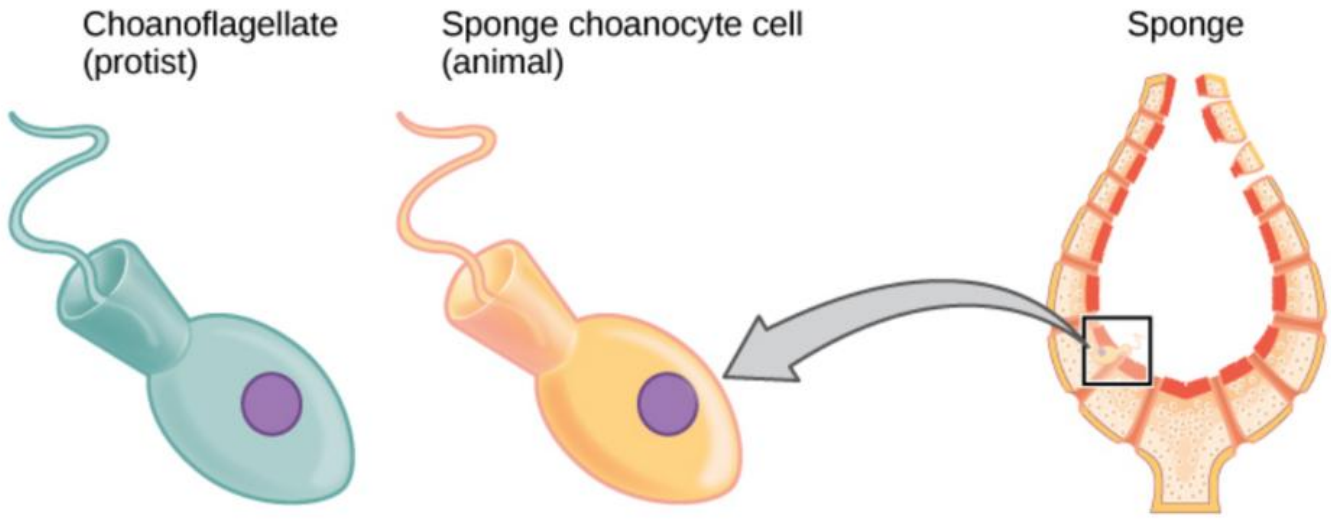
teal = contains multicellular forms; grey = unicellular. Simplified.

Key idea

Multicellular groups are separated by unicellular ones.

The closest living relatives of animals

A2.2.14



A choanoflagellate and a sponge collar cell. Clark, Choi and Douglas, CC BY 4.0.

Evidence

Single cells with a collar and flagellum, like sponge collar cells. Some form colonies; genes place them beside animals.

Discuss

Why have single cells not been replaced by multicellular organisms?

Practice: specialization

A2.2.13–14

Explain

How cells with one genome become different.

Outline

Two advantages of multicellularity.

Evaluate

Evidence that multicellularity evolved many times.

Use

gene expression, proteome, differentiation, independent

Check-in: class 2

A2.2.13–14

1. Define differentiation.
2. Explain why all body cells have the same genome.
3. Distinguish genome and proteome.
4. Give two advantages of multicellularity.
5. Explain why multicellularity is thought to have evolved several times.

How to use this

Answer without your notes, then check against slides 13–19.

Common mistakes to avoid

Not this	But this
Mitochondria came from viruses.	From aerobic bacteria.
Endosymbiosis made the nucleus.	It explains mitochondria and chloroplasts.
Specialized cells lose unused genes.	They keep them but do not express them.

Summary: cell origins and specialization

A2.2.12–14

	Key idea
A2.2.12	Endosymbiosis: mitochondria then chloroplasts
A2.2.12	Evidence: 70S, circular DNA, fission, membranes
A2.2.13	Differentiation by gene expression
A2.2.14	Multicellularity evolved many times

Check that you can...

1. **Explain** the endosymbiotic origin of organelles
2. **Evaluate** evidence for endosymbiosis
3. **Explain** differentiation
4. **Outline** the evolution of multicellularity

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State two features of mitochondria that suggest they were once bacteria.

2. Explain how cells with identical genomes become different.

3. Give one reason multicellularity is thought to have evolved more than once.

Image credits

Paramecium bursaria – 400x (13263096305).jpg – Picturepest – CC BY 2.0

Rickettsia rickettsii.jpg – CDC – Public domain

Mitochondria, mammalian lung – TEM.jpg – Louisa Howard – Public domain

Chloroplast in leaf of Anemone sp TEM 12000x.png – and3k and caper437 – CC BY-SA 3.0

Red White Blood cells.jpg – Electron Microscopy Facility at The National Cancer Institut – Public domain

Skeletal muscle – longitudinal section.jpg – Author not stated – CC BY-SA 3.0

Volvox aureus – 160x – Crosssection – daughter colonies (14811155248).jpg – Picturepest – CC BY 2.0

Choanoflagellate and choanocyte.png – Clark MA, Choi J and Douglas M – CC BY 4.0

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